

Swiss Institute of Banking and Finance



University of St.Gallen

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# **Case Study Session 1: Valuation – Consolidated Edison & Chemex AG**

**3,130/4,130 Corporate Finance**

# Key messages of this exercise

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1. The Discounted Dividend Model and the Gordon Growth Model use dividends as valuation factors.
2. The discounted cash flow (DCF) method is based on free cash flows as the respective input factors.
3. Multiples value the firm based on peer comparison.

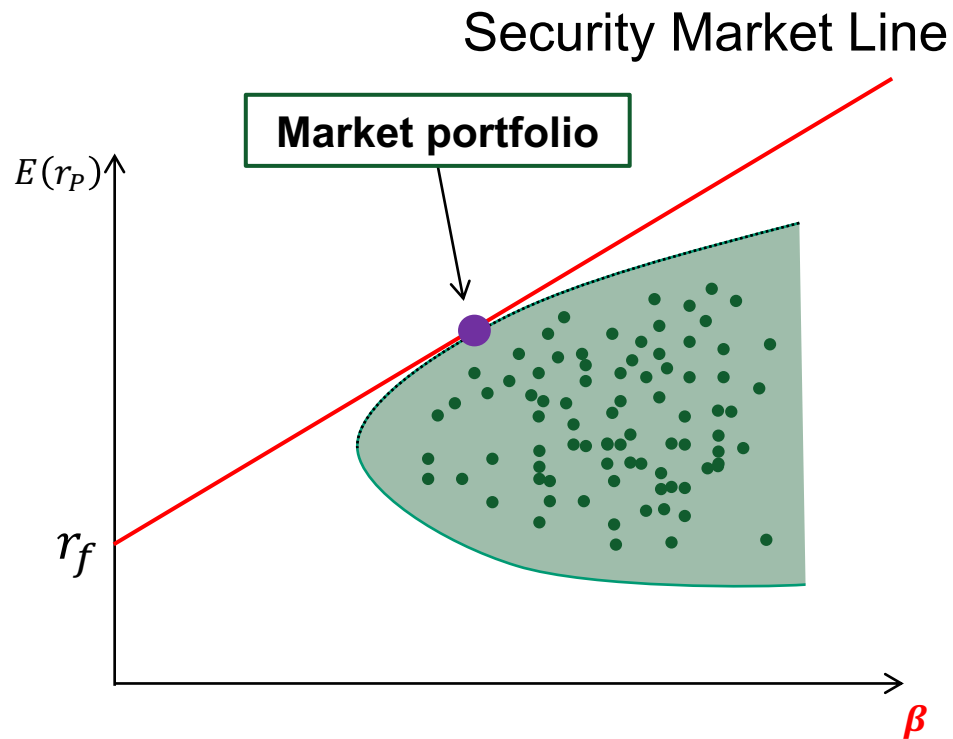
# Outline

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1. Recap: Portfolio theory
2. Consolidated Edison
3. Chemex AG

# 1. From the market portfolio to the CAPM

## 1. How are the market portfolio and beta related?



# 1. Capital Asset Pricing Model (CAPM)

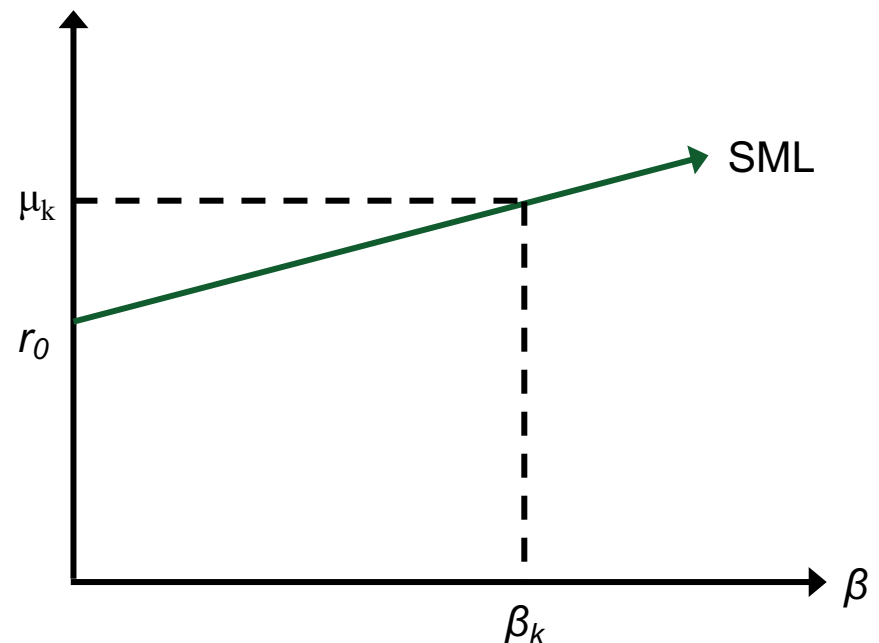
According to the CAPM, the expected return of every asset is determined (1) by the rate of interest, (2) by the systematic risk of the asset considered which is measured by beta (how the asset covaries with the market), and (3) the expected return of the market portfolio.

Formula:

$$\mu_k = r_0 + \beta_k \cdot (\mu_M - r_0)$$

## Security-Market-Line (SML)

Expected return



# 1. Beta of an asset

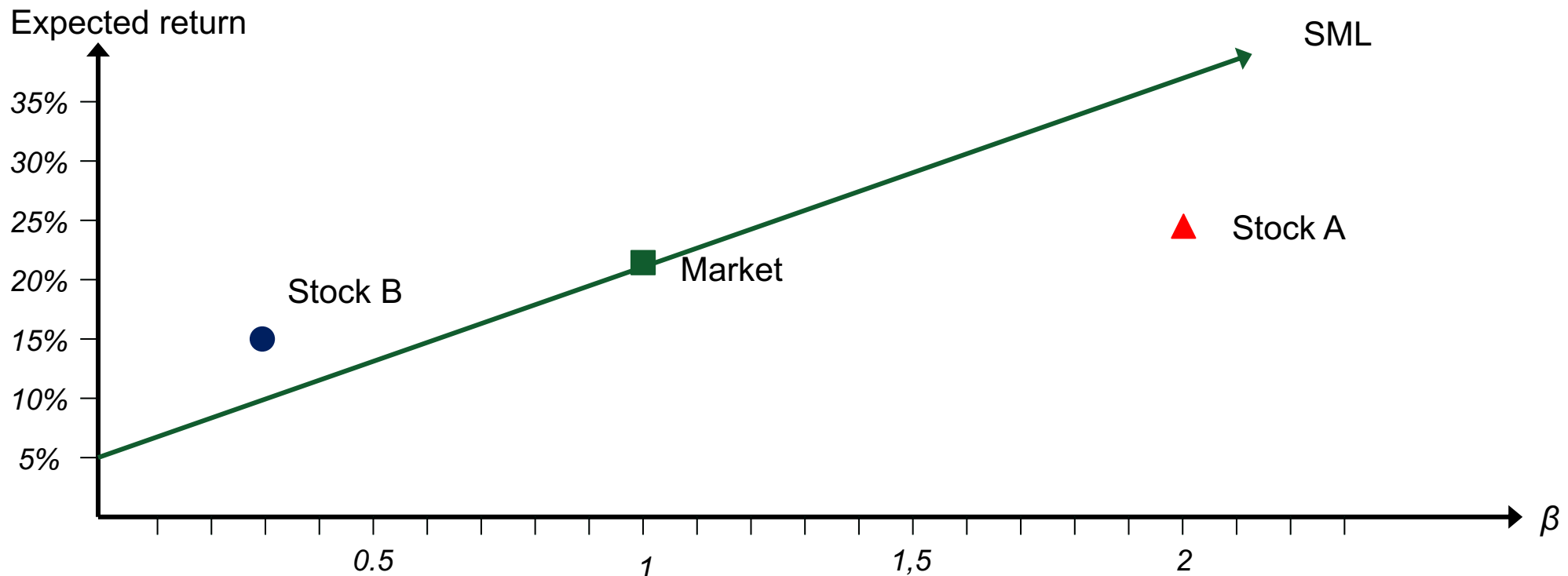
1a. What is the expected price next year according to the CAPM?

**Assumption:**

- Current stock price ( $P_0$ ) = CHF 50
- Next year dividend ( $D_1$ ) = CHF 6
- Beta = 1.2
- Market risk premium = 10%
- Risk-free interest rate = 6%

# 1. Security Market Line

1b. Suppose two stocks which lie above (Stock B) and below (Stock A) the Security Market Line (SML) respectively. What price reaction of the shares would you expect (ceteris paribus)?



## 2. Valuation of Consolidated Edison

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1. We will estimate the value of equity of Consolidated Edison applying the Dividend Discount Model.

$$P_0 = \frac{Div_1}{r}$$

2. In the next step, we will refine the model introducing the growth rate to estimate the value of equity with the Gordon Growth Model.

$$P_0 = \frac{Div_1}{r - g}$$

This case is based on the book “Investment Valuation” of Damodaran (3<sup>rd</sup> Edition).

## 2. Consolidated Edison – Corporate Profile



- For more than 180 years, Consolidated Edison, Inc. has served the world's most dynamic and demanding marketplace - metropolitan New York.
- Con Edison's principal business segments are Consolidated Edison Company of New York's regulated electric, gas and steam utility activities, Orange & Rockland Utilities' (O&R) regulated electric and gas utility activities, and Con Edison's competitive energy businesses.

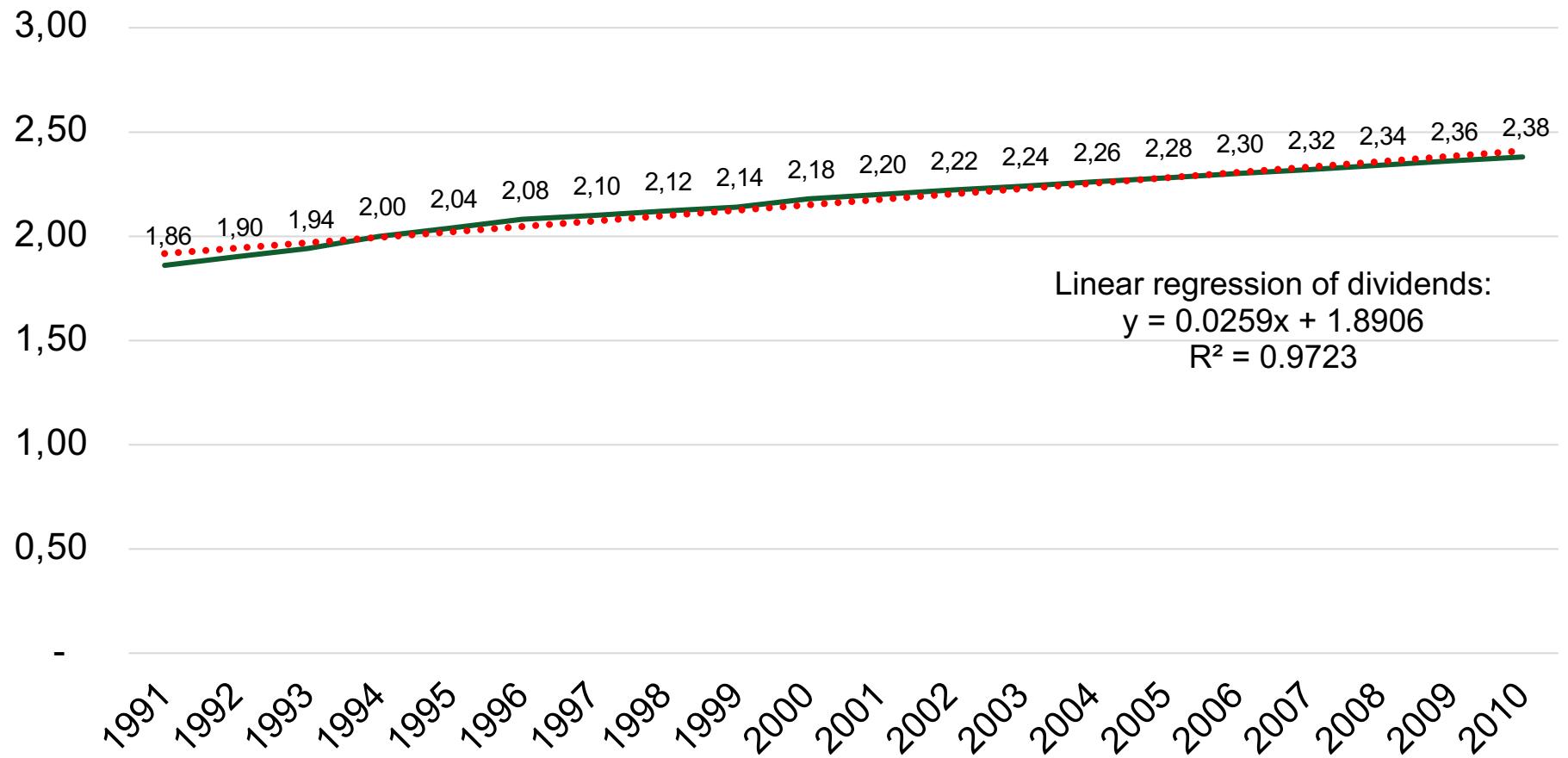
Source: "<https://www.conedison.com/en/about-us/our-businesses>"

## 2. Financial Highlights 2010

|                                    | (\$m)   |
|------------------------------------|---------|
| Operating Revenues                 | 13,325  |
| Net Income                         | 981     |
| Total Assets                       | 36,146  |
| Market Capitalization              | 14,400  |
| Capital Expenditure                | 2,029   |
|                                    |         |
| Dividends per Share                | \$2.38  |
| Stock Price per Share (Year End)   | \$49.57 |
| 5-year Annual Dividend Growth Rate | 0.86%   |
| Dividend payout ratio              | 69%     |

Source: Annual Report 2010 and Website

## 2. Yearly Dividend Payments per Share 1991-2010



Source: Consolidated Edison – Dividend History

## 2. Cost of Equity 2006-2010

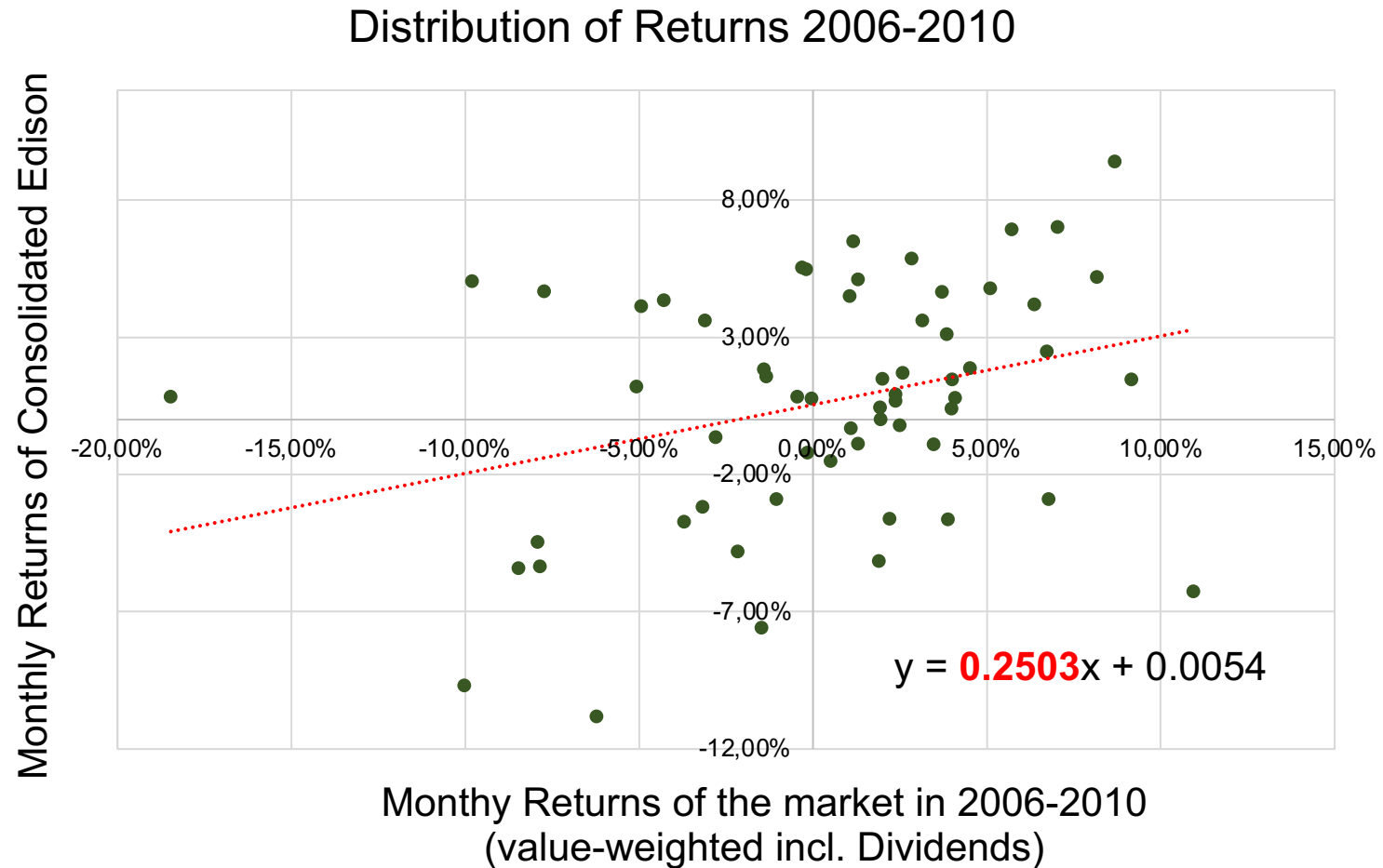
2a. Calculate the cost of equity applying the CAPM!

Risk-free interest rate = 3.91%

Market risk premium = 4.60%

Beta = 0.25

## 2. Intuition of the Beta 2006-2010



Source: CRSP

## 2. Dividend Discount Model 2011

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2b. Calculate reasonable values for the expected next year dividend!

## 2. Dividend Discount Model 2011

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2c. What would be the estimated stock price at the beginning of 2011 ?

## 2. Gordon Growth Model 2011

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2d. What would be the estimated stock price at the beginning of 2011 if you also consider the growth of the dividends?

## 2. Gordon Growth Model

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2e. Why does the Gordon Growth Model perform fairly well in this situation?

## 2. More recent past of Consolidated Edison

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- We now look into more recent data to compute the price per share of Consolidated Edison using the Gordon Growth Model and test whether the model still holds.
- Does the Gordon Growth Model work well in the more recent past?
  - Are there any changes in the recent past which make it harder apply the Gordon Growth Model?
  - We will have a look into Consolidated Edison's dividend payments.
  - We will recalculate/ reestimate Consolidated Edison's cost of equity.

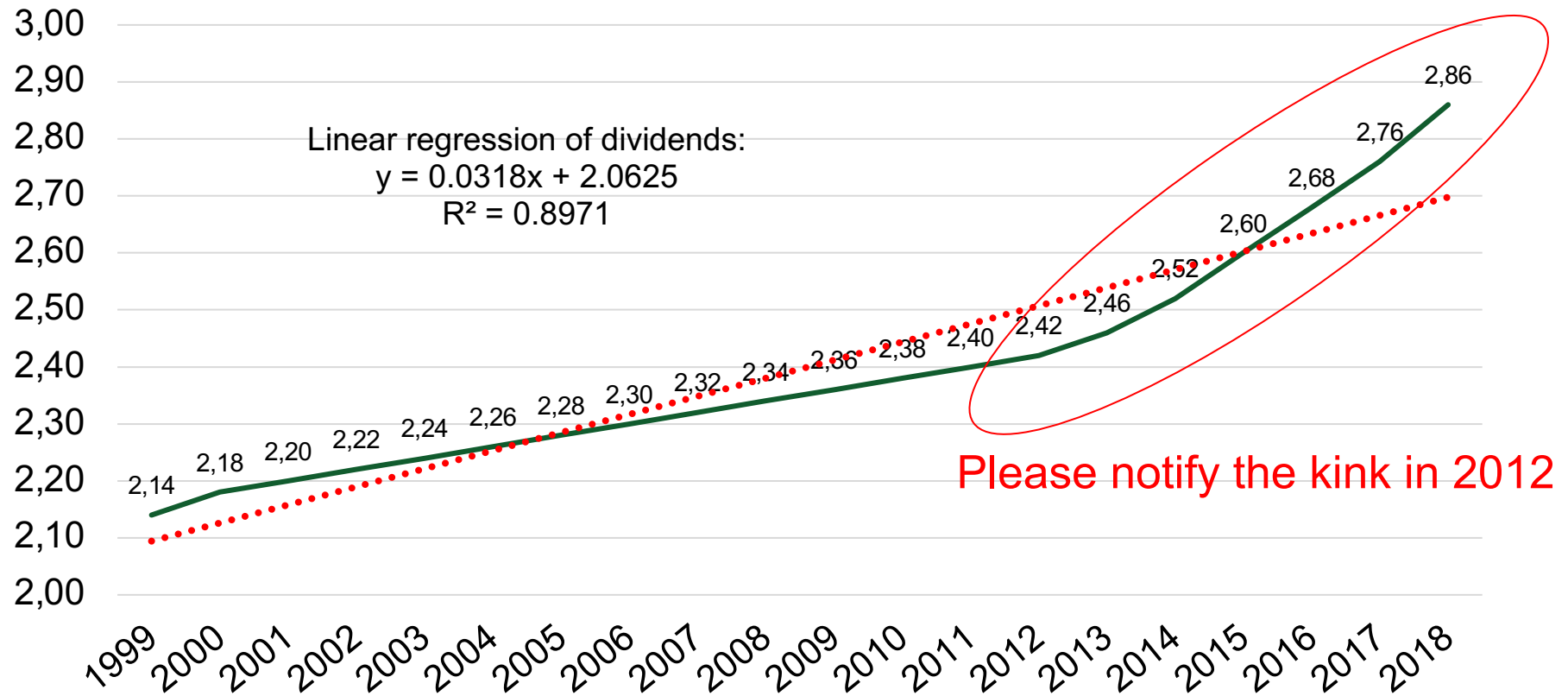
## 2. Financial Highlights 2018

|                                    | (\$m)        |
|------------------------------------|--------------|
| Operating Revenues                 | 12,337       |
| Net Income                         | 1,382        |
| Total Assets                       | 53,920       |
| Market Capitalization              | 24,500       |
| Capital Expenditure                | 5,249        |
|                                    |              |
| Dividends per Share                | \$2.86       |
| Stock Price per Share (Year End)   | \$76.46      |
| 5-year Annual Dividend Growth Rate | <b>3.06%</b> |
| Dividend payout ratio              | 65%          |

vs. 0.86% in 2010

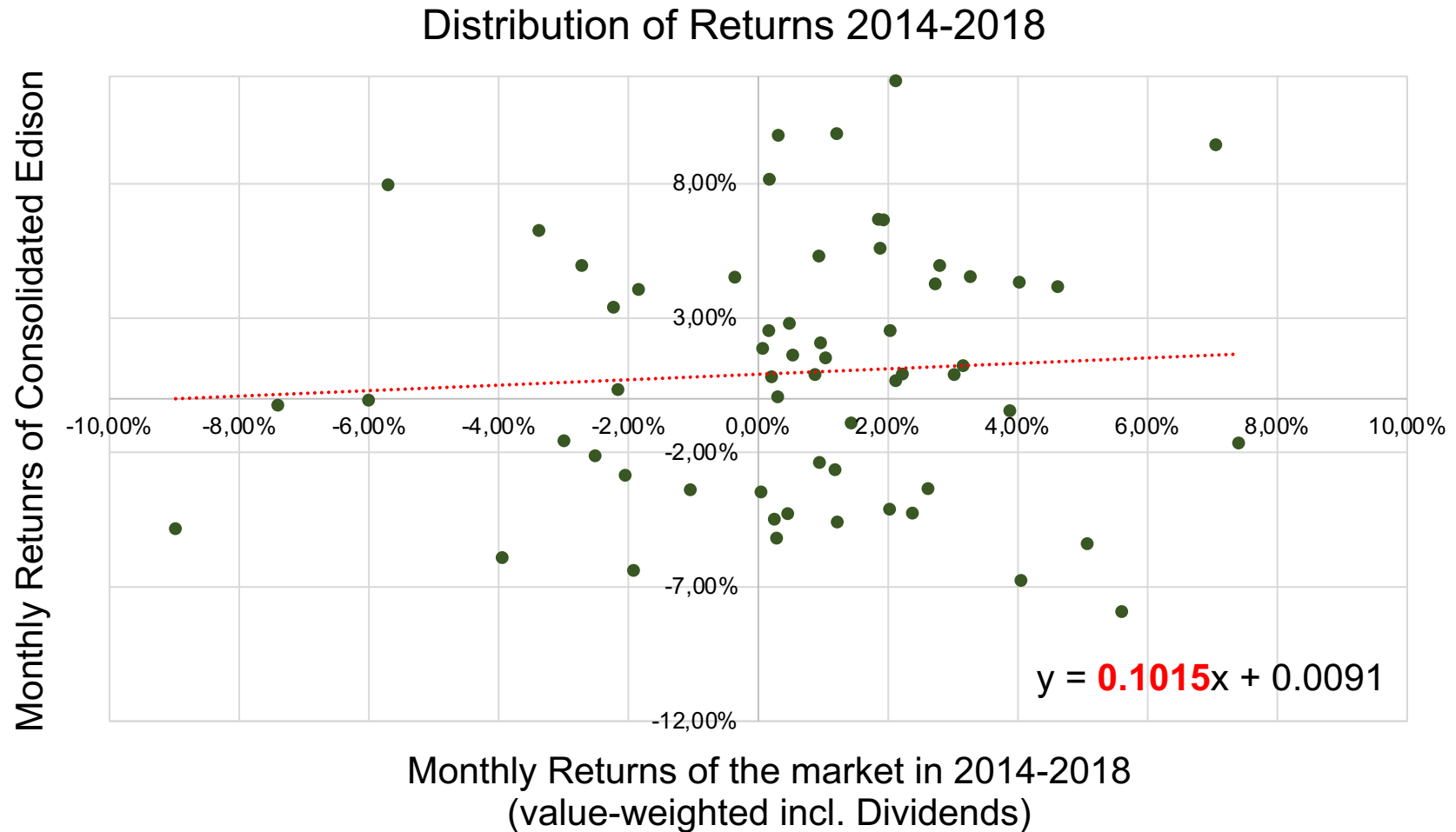
Source: Annual Report 2018 and Website

## 2. Yearly Dividend Payments per Share 1999-2018



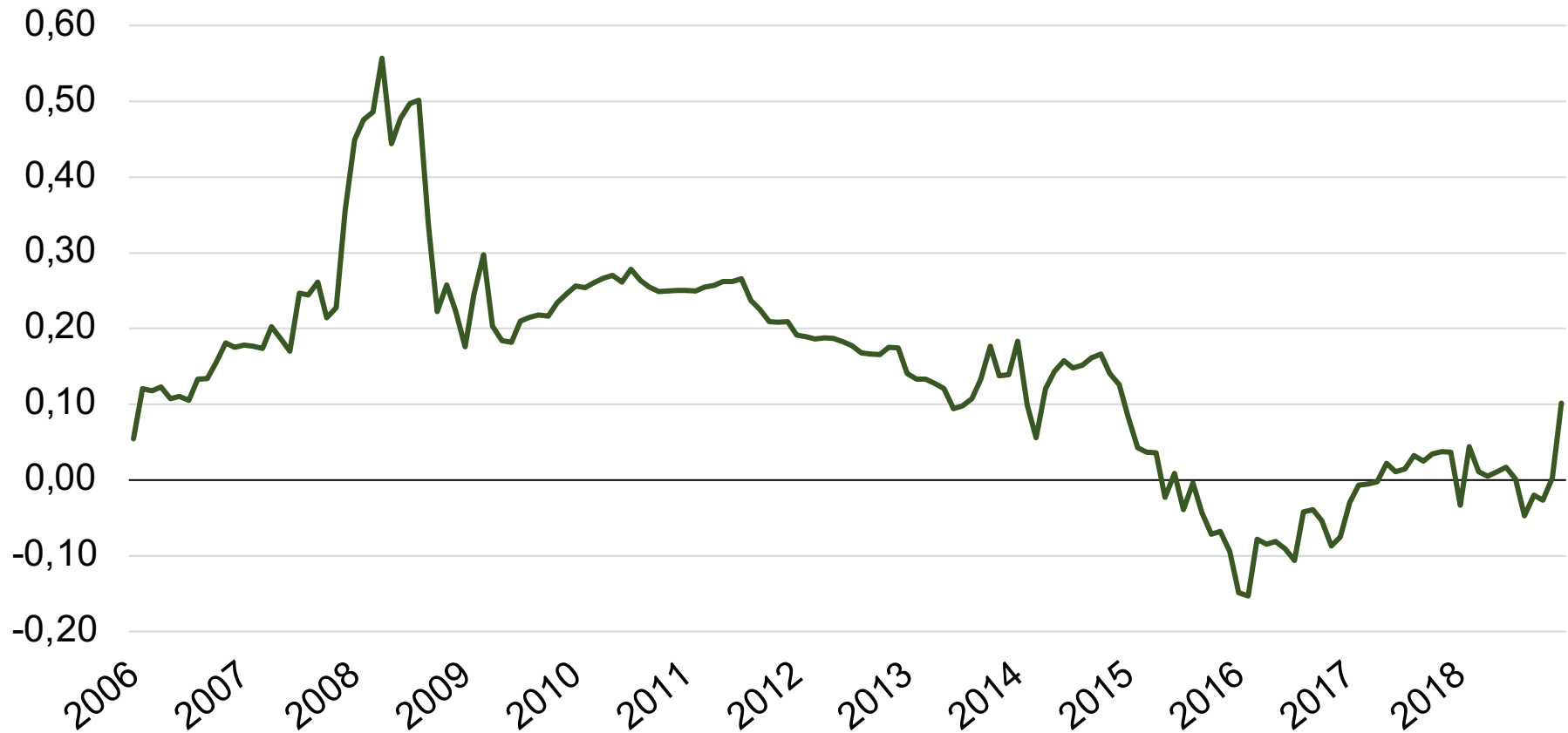
Source: Consolidated Edison – Dividend History

## 2. Intuition of the Beta 2014-2018



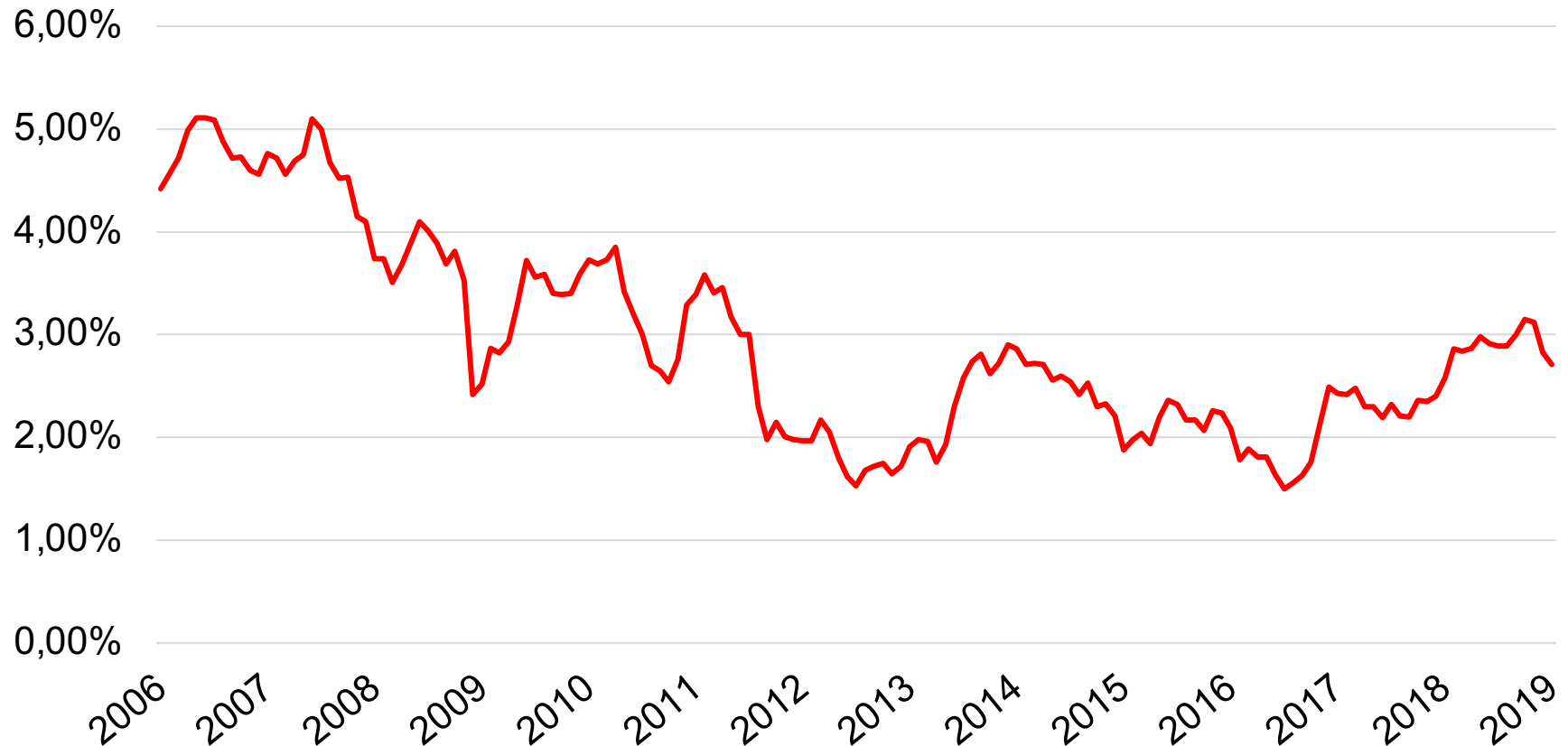
Source: CRSP

## 2. Five-year Rolling Monthly Beta 2006-2018



Source: CRSP

## 2. Monthly Risk-free Rate 2006-2018



Source: <https://fred.stlouisfed.org/series/GS10>

## 2. Cost of Equity 2014-2018

2f. Calculate the cost of equity applying the CAPM!

Risk-free interest rate = 2.35%

Market risk premium = 4.19%

Beta = 0.10

## 2. Dividend Discount Model 2019

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2g. Calculate reasonable values for the expected next year dividend!

## 2. Gordon Growth Model 2019

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2h. What would be the estimated stock price at the beginning of 2019 if you also consider the growth of the dividends?

## 2. Discounted Cash Flow (DCF) Model

- Instead of using dividends in the Dividend Discount Model (DDM), free cash flows (FCF) can also be used to **value a company**.
- The value of a business is usually computed as the discounted value of free cash flows out to a valuation horizon ( $H$ ), plus the forecasted value of the business  $PV_H$  at this horizon, also discounted back to present value.

$$PV_0 = \frac{FCF_1}{1+r} + \frac{FCF_1}{(1+r)^2} + \dots + \frac{FCF_H}{(1+r)^H} + \frac{PV_H}{(1+r)^H} = \sum_{t=1}^H \frac{FCF_t}{(1+r)^t} + \frac{PV_H}{(1+r)^H}$$

Sum of present values of the  
free cash flows planned **in detail**  
for the coming  $H$  years

Discounted  
horizon  
value

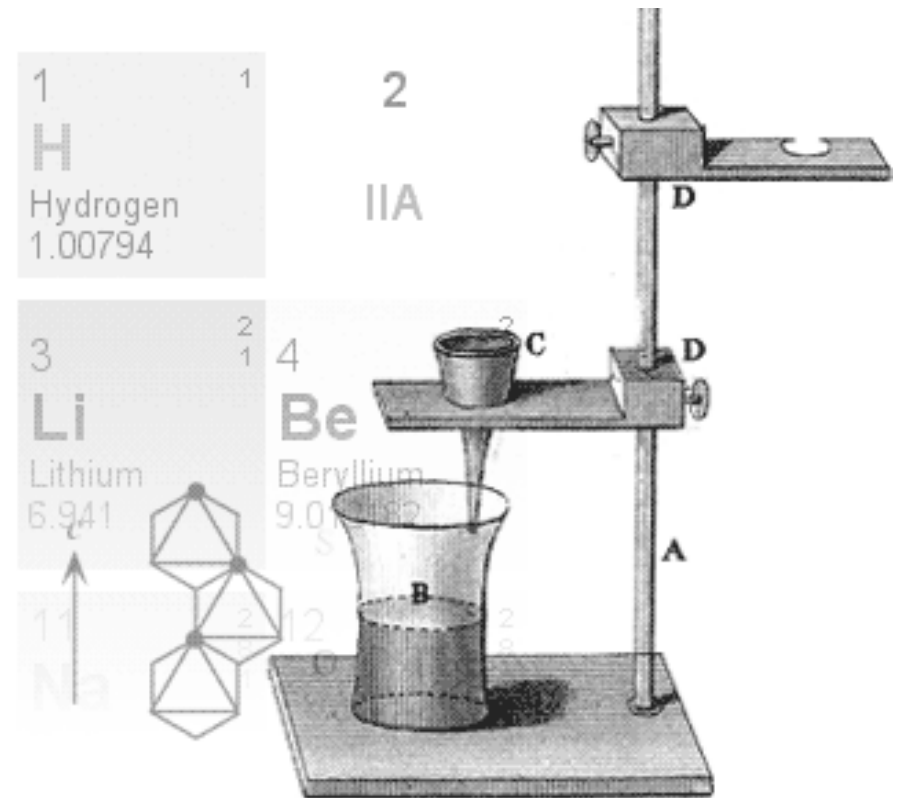
### 3. «Chemex AG»

Chemex AG is a leading international company in the specialty chemicals sector with a strong customer focus and innovative products.

Its market segments include:

- Textiles, leather & paper chemicals
- Pigments and chem. additives
- Monomers and water-soluble polymers
- Life-science chemicals

A private equity firm is attracted by the long-term operational performance and thus interested in buying the Chemex AG.



### 3. Condensed Annual Report 2018

#### Balance Sheet 2018

| <b>Assets</b>                         | <b>(\$m)</b> |
|---------------------------------------|--------------|
| Cash (50% excess liquidity)           | 550          |
| Receivables                           | 1'100        |
| Inventories                           | 1'300        |
| Plan, property, equip. (PPE)          | 2'400        |
|                                       | <b>5'350</b> |
| <b>Liabilities</b>                    | <b>(\$m)</b> |
| Accounts payable                      | 1'200        |
| Bank loans                            | 50           |
| Long-term debt                        | 2'100        |
| Equity* (incl. net income – dividend) | 2'000        |
|                                       | <b>5'350</b> |

#### Income Statement 2018

|                                                                        |            |
|------------------------------------------------------------------------|------------|
| Sales                                                                  | 8'500      |
| - Cost of sales                                                        | -7'600     |
| <b>EBITDA</b>                                                          | <b>900</b> |
| <i>Earnings before interests, taxes, depreciation and amortization</i> |            |
| - Depreciation                                                         | -340       |
| <b>EBIT</b>                                                            | <b>560</b> |
| <i>Earnings before interests and taxes</i>                             |            |
| - Interest expenses                                                    | -127       |
| <b>EBT</b>                                                             | <b>433</b> |
| <i>Earnings before taxes</i>                                           |            |
| - Taxes                                                                | -108       |
| <b>Earnings after taxes/ Net income</b>                                | <b>325</b> |

\*100m shares outstanding

### 3. Forecast Investment Plan

Assume that payments for investments occur at the end of each year.

|                          | 2019  | 2020  | 2021  | 2022  | 2023  | 2024           |
|--------------------------|-------|-------|-------|-------|-------|----------------|
| Plant, property & equip. | 2'400 | 2'412 | 2'453 | 2'509 | 2'538 | 2'558          |
| CAPEX                    | 300   | 330   | 350   | 330   | 325   | = depreciation |
| Depreciation             | -288  | -289  | -294  | -301  | -305  | -307           |
| Production equipment     | 2'412 | 2'453 | 2'509 | 2'538 | 2'558 | 2'558          |

### 3. Forecast Income Statement

|                                 | %   | 2019       | 2020       | 2021       | 2022       | 2023       | 2024       |
|---------------------------------|-----|------------|------------|------------|------------|------------|------------|
| Sales                           |     | 8'755      | 9'018      | 9'198      | 9'382      | 9'476      | 9'571      |
| <i>Growth forecast in sales</i> |     | 3%         | 3%         | 2%         | 2%         | 1%         | 1%         |
| - Cost of goods sold            |     | -7'880     | -8'116     | -8'278     | -8'444     | -8'528     | -8'614     |
| <b>EBITDA</b>                   |     | <b>875</b> | <b>902</b> | <b>920</b> | <b>938</b> | <b>948</b> | <b>957</b> |
| - Depreciation                  |     | -288       | -289       | -294       | -301       | -305       | -307       |
| <b>EBIT</b>                     |     | <b>587</b> | <b>613</b> | <b>626</b> | <b>637</b> | <b>643</b> | <b>650</b> |
| - Interests                     |     | -130       | -133       | -135       | -137       | -139       | -142       |
| <b>EBT</b>                      |     | <b>457</b> | <b>480</b> | <b>491</b> | <b>500</b> | <b>504</b> | <b>508</b> |
| - Tax                           | 25% | -114       | -120       | -123       | -125       | -126       | -127       |
| <b>Net income</b>               |     | <b>343</b> | <b>360</b> | <b>368</b> | <b>375</b> | <b>378</b> | <b>381</b> |
| Dividend                        | 90% | 309        | 324        | 331        | 338        | 340        | 343        |

### 3. Forecast Balance Sheet

| Assets (in \$ million)                | 2019         | 2020         | 2021         | 2022         | 2023         | 2024         |
|---------------------------------------|--------------|--------------|--------------|--------------|--------------|--------------|
| Cash & Equivalents                    | 424          | 418          | 407          | 430          | 473          | 537          |
| Accounts receivable                   | 1'151        | 1'186        | 1'210        | 1'234        | 1'246        | 1'259        |
| Inventories                           | 1'313        | 1'353        | 1'380        | 1'407        | 1'421        | 1'436        |
| Plant, properties & equip.            | 2'412        | 2'453        | 2'509        | 2'538        | 2'558        | 2'558        |
| (CAPEX as part of PPE)                | (300)        | (330)        | (350)        | (330)        | (325)        | (307)        |
| <b>Total Assets</b>                   | <b>5'300</b> | <b>5'410</b> | <b>5'506</b> | <b>5'609</b> | <b>5'698</b> | <b>5'790</b> |
| Liabilities (in \$ million)           | 2019         | 2020         | 2021         | 2022         | 2023         | 2024         |
| Accounts payable                      | 1'079        | 1'112        | 1'134        | 1'157        | 1'168        | 1'180        |
| Short term debt                       | 37           | 27           | 15           | 32           | 47           | 38           |
| Long term debt                        | 2'150        | 2'200        | 2'250        | 2'275        | 2'300        | 2'350        |
| Equity                                | 2'034        | 2'071        | 2'107        | 2'145        | 2'183        | 2'222        |
| <b>Total Liabilities &amp; Equity</b> | <b>5'300</b> | <b>5'410</b> | <b>5'506</b> | <b>5'609</b> | <b>5'698</b> | <b>5'790</b> |

# 3. Free Cash Flows

## Free cash flow:

Cash flows created by the firm which are available for shareholders and creditors.

## Free cash flow calculation

EBIT

- Taxes on EBIT

**NOPAT**

*Net operating profit after taxes*

+ Depreciation

**Cash flow**

- *increase of working capital*

- CAPEX

**Free cash flow**

### 3. Forecast the free cash flows

|                                                | 2019       | 2020       | 2021       | 2022       | 2023 | 2024 |
|------------------------------------------------|------------|------------|------------|------------|------|------|
| EBIT                                           | 587        | 613        | 626        | 637        |      |      |
| - Taxes                                        | -147       | -153       | -157       | -159       |      |      |
| <b>NOPAT</b>                                   | <b>440</b> | <b>460</b> | <b>469</b> | <b>478</b> |      |      |
| + Depreciation                                 | 288        | 289        | 294        | 301        |      |      |
| Change in working capital                      | -185       | -42        | -29        | -28        |      |      |
| CAPEX                                          | -300       | -330       | -350       | -330       |      |      |
| <b>Free cash flow</b>                          | <b>243</b> | <b>377</b> | <b>384</b> | <b>421</b> |      |      |
| <b>Calculate the change in working capital</b> |            |            |            |            |      |      |
| Increase in receivables                        | 51         | 35         | 24         | 24         |      |      |
| Increase in inventory                          | 13         | 40         | 27         | 27         |      |      |
| - Increase in payables                         | 121        | -33        | -22        | -23        |      |      |
| <b>Increase in working capital</b>             | <b>185</b> | <b>42</b>  | <b>29</b>  | <b>28</b>  |      |      |

### 3. Parts of the company value

| Company value                                                      |                                                                                   |                                                                                                                                                                                                                                                                                |
|--------------------------------------------------------------------|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Forecasting horizon                                                |                                                                                   | Horizon value                                                                                                                                                                                                                                                                  |
| Sum of the discounted free cash flows over the forecasting horizon |  | <p>The 'horizon value' is a lump-sum estimation of the <i>present value at a future point in time</i> of all subsequent future cash flows (typically assumed to grow perpetually at a constant rate).</p> $\text{Horizon value}_T = \frac{FCF_{T+1}}{\text{capital cost} - g}$ |

### 3. Forecast the horizon value

Calculate the horizon value of Chemex AG

Assumptions:

Cost of capital = 8.5%

Organic growth rate = 2.0%

Horizon value<sub>2024</sub> =

### 3. Enterprise and equity value

Calculate the enterprise and equity value!

#### Operating result (PV of FCF)

- + excess liquidity
- + investment in unconsolidated subsidiaries
- + other non-operative asset values

#### Enterprise value

- interest bearing debt
- capitalized value of operative leasing
- minority interests
- preferred stocks
- employee stock options

#### Equity value

Note: Preferred stocks are treated as equity in other legal systems (e.g. Germany).

### 3. Enterprise and equity value

| DCF value of the company      | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 |
|-------------------------------|------|------|------|------|------|------|
| Free cash flows               |      |      |      |      |      |      |
| Horizon value (g = 2%)        |      |      |      |      |      |      |
| Discount factor (WACC = 8.5%) |      |      |      |      |      |      |
| Discounted free cash flows    |      |      |      |      |      |      |
| Discounted horizon value      |      |      |      |      |      |      |

|                                       |  |
|---------------------------------------|--|
| Sum of discounted free cash flows     |  |
| Discounted horizon value              |  |
| <b>Operative value of the company</b> |  |
| + excess liquidity                    |  |
| <b>Enterprise value</b>               |  |
| - interest bearing debt               |  |
| <b>Equity value</b>                   |  |

### 3. Impact of CAPEX

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How does the value of the enterprise and the equity change if the CAPX will be reduced by 50 every year?

### 3. Enterprise and equity value

| DCF value of the company      | 2019   | 2020   | 2021   | 2022   | 2023   | 2024   |
|-------------------------------|--------|--------|--------|--------|--------|--------|
| Free cash flows               |        |        |        |        |        |        |
| Change in CAPEX               |        |        |        |        |        |        |
| Horizon value (g = 2%)        |        |        |        |        |        |        |
| Discount factor (WACC = 8.5%) | 0.9217 | 0.8495 | 0.7829 | 0.7216 | 0.6650 | 0.6129 |
| Discounted free cash flows    |        |        |        |        |        |        |
| Discounted horizon value      |        |        |        |        |        |        |

|                                       |  |
|---------------------------------------|--|
| Sum of discounted free cash flows     |  |
| Discounted horizon value              |  |
| <b>Operative value of the company</b> |  |
| + excess liquidity                    |  |
| <b>Enterprise value</b>               |  |
| - interest bearing debt               |  |
| <b>Equity value</b>                   |  |

### 3. Interpretation of the results

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Do lower CAPEX really increases the firm value?

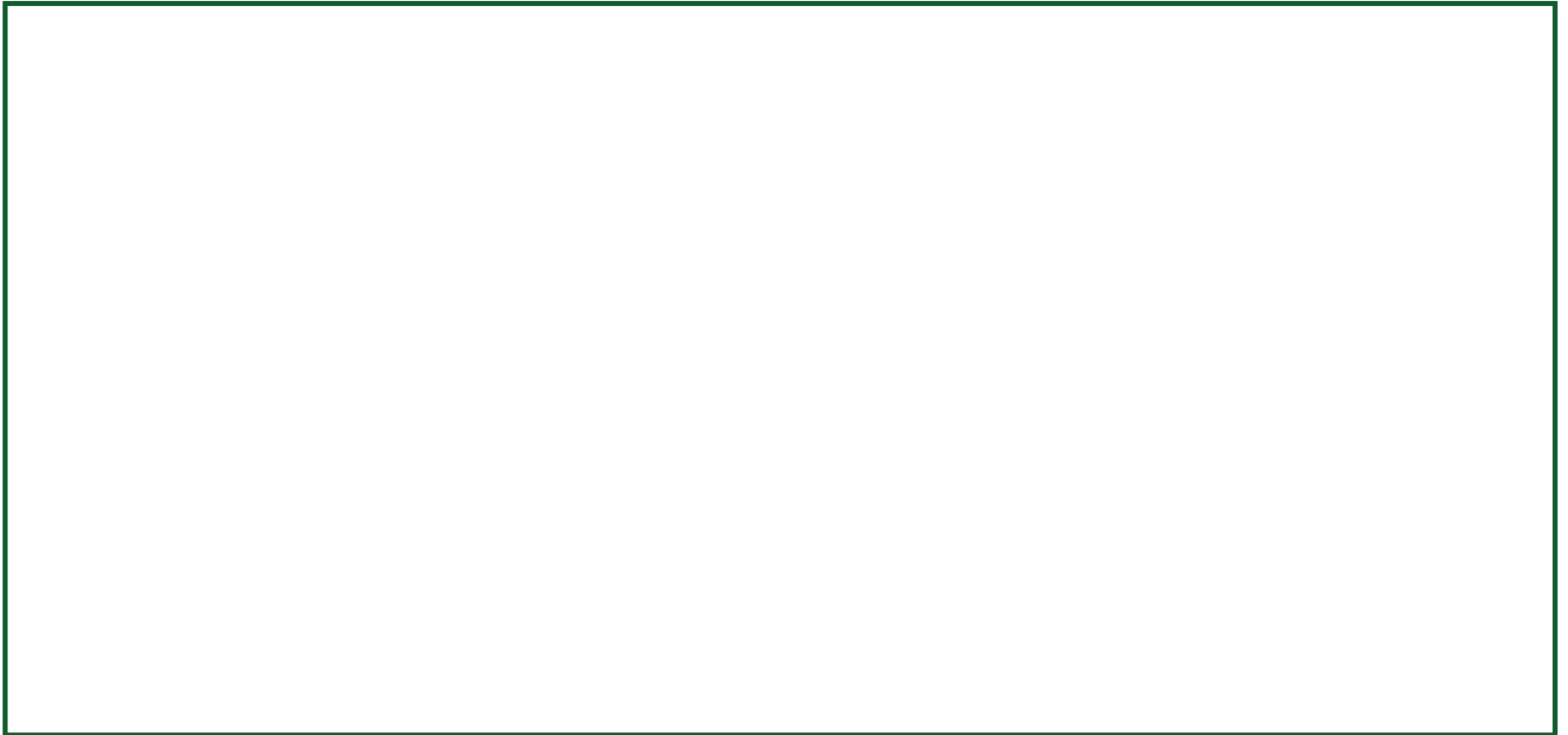
### 3. Evaluation of DCF

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Assess the advantages and drawbacks of the DCF valuation method!

### 3. Advantages of DCF

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### 3. Disadvantages of DCF

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# 3. Multiple Valuation

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How does the multiple valuation method work?

What are advantages and drawbacks?

# 3. Multiple Valuation

## Multiple valuation

- **Concept:**
- **Construction:**
- **Categories:**
- **Approach:**
  - 1.
  - 2.
  - 3.
  - 4.
  - 5.

### 3. Multiple Valuation

| Multiple valuation |           |
|--------------------|-----------|
| Advantages         | Drawbacks |
| Hints              |           |

### 3. Multiple Valuation

Suppose the following peers:

| Peer companies         | Last Price | P/E Forecast FYR1 | Exp. growth rate |
|------------------------|------------|-------------------|------------------|
| TeChem Ltd.            | 32.41      | 10.9              | 1.9%             |
| Chem Specialty SA      | 49.04      | 13.2              | 2.2%             |
| Chemical & Brothers AG | 11.03      | 14.1              | 2.1%             |
| SynChem, Inc.          | 547.25     | 9.4               | 1.2%             |
| CSP Co                 | 48.52      | 9.8               | 1.3%             |
| Sinochemicals, Inc.    | 101.80     | 10.6              | 1.8%             |
| AirChemicals Co        | 13.91      | 12.5              | 2.0%             |
| <b>Average</b>         |            | <b>11.5</b>       | <b>1.8%</b>      |
| <b>Median</b>          |            | <b>10.9</b>       | <b>1.9%</b>      |

### 3. Multiple Valuation

Please estimate the value of Chemex AG using the P/E ratios!

| P/E ratio |
|-----------|
|-----------|

### 3. Multiple Valuation

What is the P/E ratio considering the DCF valuation?  
Is the company over- or undervalued compared with its peers?

**Implied P/E ratio**

### 3. Multiple Valuation

Is Chemex over- or undervalued if the growth rates are considered?

| Peer companies         | P/E         | g           | PEG* |
|------------------------|-------------|-------------|------|
| TeChem Ltd.            | 10.9        | 1.9%        |      |
| Chem Specialty SA      | 13.2        | 2.2%        |      |
| Chemical & Brothers AG | 14.1        | 2.1%        |      |
| SynChem, Inc.          | 9.4         | 1.2%        |      |
| CSP Co                 | 9.8         | 1.3%        |      |
| Sinochemicals, Inc.    | 10.6        | 1.8%        |      |
| AirChemicals Co        | 12.5        | 2.0%        |      |
| <b>Average</b>         | <b>11.5</b> | <b>1.8%</b> |      |
| <b>Median</b>          | <b>10.9</b> | <b>1.9%</b> |      |

\* growth-adjusted P/E

### 3. Multiple Valuation

Is Chemex over- or undervalued if the growth rates are considered?

#### PEG ratio of Chemex

# Recap

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- The Dividend Discount Model and the Gordon Growth Model may perform well if the environment of the firm and the firm itself are relatively stable.
- The Discounted Cash Flow Model is based on the Free Cash Flows of a firm so that it is independent of any accounting or dividend policies. However, the model is highly sensitive to the several estimation parameters (e.g. terminal growth).
- In contrast to the DDM, GGM, and the DCF, multiple valuation uses comparable firms or transactions to estimate the value. Therefore, the “market mood” will be reflected.

# Literature

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## **Additional literature**

- Damodaran, A., 2010, The Dark Side of Valuation, 2nd edition, Prentice Hall.
- Koller, T., Goedhart, M., and D. Wessels, 2010, Valuation: Measuring and Managing the Value of Companies, 5th edition, John Wiley & Sons.
- Palepu, K., and P. Healy, 2012, Business Analysis & Valuation: Using Financial Statements, 5th edition, South-Western Education Publishing.